Skylark Intelligence - Decision Log

Monday.com Business Intelligence Agent | Final implementation
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1. Product interpretation and scope

I interpreted the assignment as a founder-facing intelligence layer over live Monday.com data, not as a general-purpose chatbot. The required evaluation path remains deliberately optimized for the two supplied private boards - **Deals** and **Work Orders** - while an optional **Analyze your own boards** mode extends the prototype to one or more arbitrary Monday boards without changing the assignment-specific workflow.

Core outcome

Users can ask conversational questions about pipeline, revenue, sector performance, delivery status, collections, operational risk and data quality. The application reads Monday dynamically at request time, computes reliable metrics in code, and uses the language model tointerpretquestions and communicate executive-level findings.

2. Key assumptions

- **Two-board compliance:**Theassignment requires complete read access to the supplied Deals and Work Orders boards; it does not require unrestricted workspace access.
- **Read-only means no mutations:** The prototype does not create, edit or delete Monday items. Board setup/import is performed separately by the account owner.
- **Messy data is expected:** Missing dates, inconsistent labels, formatted numbers, duplicate-looking values and incomplete records are normal inputs, not fatal errors.
- **Business answers require caveats:** Where a field is missing or an inferred mapping is uncertain, the response should disclose the limitation rather than fabricate precision.
- **No persistent identity or history wasrequired:**Thehosted prototype is intentionally usable without accounts or a database.

3. Architecture and technology choices

Next.js + TypeScript on Vercel. This providedasingledeployable codebase for the responsive interface and server-side API routes, strong typing, fast iteration, and a zero-local-setup hosted prototype.

Monday GraphQL API. Direct API integration was chosen for explicit read-only queries, pagination control, predictable deployment and transparent error handling. Board IDs are configuration rather than hardcoded CSV content.

Hugging Face inference. A fine-grained Hugging Face token and hosted instruction model were used instead of requiring a paid OpenAI API key. This reduced setup cost while preserving conversational query interpretation. The provider/model boundary is isolated so it can be replaced later.

Hybrid BI pipeline. Dates, currencies, percentages, totals, averages, counts and status distributions are normalized and calculated deterministically. AI is used for intent interpretation, schema selection, explanation and follow-up insight - not as the sole calculator.

Decision principle

Reliability was prioritized over maximum autonomy: deterministic calculations remain auditable, while model-generated language adds context without becoming the source of truth.

4. Data resilience and arbitrary-board mode

Thenormalization layer preserves sourceboard, itemandcolumnprovenance while handling nulls, whitespace, mixed capitalization, formatted numbers, percentages and inconsistent date representations. Assignment mode applies known Deals/Work Orders aliases for higher accuracy.

The optional arbitrary-board mode accepts one or more board IDs/URLs, discovers board names and schemas, and infers semantic roles such as status, date, amount, percentage, owner and category. Inference confidence and data-quality caveats are retained. Large-board safeguards use pagination, bounded data retrieval and question-relevant aggregation so raw datasets are not blindly sent to the model.

5. Security and privacy decisions

- Deploymentcredentialsremainserver-sideVercelenvironment variables: Monday token, assignment board IDs and Hugging Face token.
- Assignment boards are private, and the Monday personal token inherits the owner's existing permissions.
- Temporary custom-board credentials live only in browser memory, are sent over HTTPS for the current request, are never logged or forwarded to Hugging Face, and disappear on refresh.
- Chat history is session-scoped and intentionally not stored. Only the non-sensitive light/dark/system theme preference may persist locally.
- API errors are sanitized so credentials and authorization headers cannot appear in responses.

6. User experience decisions

Thefinalinterfacewas redesigned as a premium executive SaaS workspace rather than a generic chat window. It includes responsive navigation, accessible dialogs, data-source status, suggested founder questions, structured intelligence responses, loading/error states, and polished light, dark and system themes. The theme preference persists; business data, questions and credentials do not.

I retained a specialized default experience because it provides the strongest assignment demonstration, while custom-board mode proves that the platform can adapt to unfamiliar schemas. The UI explicitly communicates when temporary boards are active and allows the user toreturn to demo data.

7. Interpretation of 'leadership updates'

I interpreted this as converting analysis in to a concise executive briefing rather than merely exporting raw data. A leadership-ready response emphasizes: headline KPIs, material changes, risks/blockers, opportunities, data-quality caveats and recommended follow-up questions. It is designed to be copied into a weekly leadership note while retaining board provenance and avoiding unsupported claims.

8. Trade-offs

- **Personal token over OAuth:** faster and appropriate for a six-hour private prototype; OAuth is the correct production multi-user design.
- **No database:** improves privacy and delivery speed, but conversations do not survive refresh and there is no authenticated history or audit trail.
- **Inference over universal ontology:** arbitrary boards are flexible, but unfamiliar or poorly labelled schemas can reduce confidence; the product asks for clarification and reports uncertainty.
- **Bounded context:** protecting latency, cost and model context can require sampling or aggregation on very large boards; responses disclose coverage limits.

9. What I would do with more time

I would add Monday OAuth with least-privilege scopes, authenticated organizations and role-based access; encrypted conversation history with retention/deletion controls; a metadata catalog with user-confirmed semantic mappings; background caching and incremental sync; richer deterministic query planning; exportable leadership briefs; observability with secret redaction; and an evaluation suite measuring numeric accuracy, schema inference, hallucination resistance and latency across many real-world boards.

Final position

The implementation meets the required two-board, read-only, dynamically queried Monday.com use case, while the optional arbitrary-board mode, privacy controls and premium themed interface demonstrate a credible path from technical assignment to commercial product.

Stack: Next.js, TypeScript, Monday GraphQL API, Hugging Face inference, Vercel.

10. Clarifying Questions and Response Accuracy

The agent may ask clarifying questions when a business query is ambiguous, references an unidentified metric or timeframe, or when connected boards contain unfamiliar or conflicting schemas. This was an intentional accuracy-first decision: answering immediately under these conditions could produce misleading calculations or unsupported conclusions. Clarification is especially important in arbitrary-board mode, where terms such as “performance,” “value,” “pipeline,” or “this quarter” may correspond to multiple columns.

However, the current prototype can occasionally ask more questions than necessary. This is a trade-off between conversational speed and preventing hallucinated assumptions. With more time, I would introduce a confidence threshold that allows the agent to proceed using clearly stated assumptions when confidence is high, ask only one consolidated clarification when uncertainty is material, and provide a preliminary answer alongside the clarification whenever possible.